

5.3.2 Data Protection (Encryption, Passwords, Backup, Firewall)

ڈیٹا پروٹیکشن کا مطلب ہے معلومات کو غیر مجاز رسائی، غلط استعمال، نقصان یا حادثاتی حذف سے بچانا۔

CS XII

Chapter 5: Application and Impacts of Computing

Lecture 09 of 15

5.3.2 Data Protection (Encryption, Passwords, Backup, Firewall)

LEARN IT

Data protection means keeping digital information safe from **unauthorised access, misuse, loss, damage or accidental deletion**.

1. Data encryption

Encryption changes readable data (**plaintext**) into a coded form called **ciphertext**.

Unauthorised users cannot understand ciphertext without the key.

- **Symmetric encryption** — the **same key** is used to encrypt and to decrypt.
- **Asymmetric encryption** — **two keys**: a **public key** to encrypt, a **private key** to decrypt.

A simple exam cipher is the **Caesar cipher** (a symmetric substitution cipher). You shift every letter a fixed number of places along the alphabet. Shift 3 is the classic key: A → D, B → E, ..., Z → C (the alphabet wraps around).

2. Strong passwords

A strong password includes **uppercase, lowercase, numbers and symbols**. It resists **brute-force** guessing (trying every possible combination). Source example: `Zk@2026#Pass`.

Weak: `ahmed123`, `pakistan`, `sindh`, your date of birth. Never reuse the same password on the college LMS and on JazzCash.

3. Backup and recovery — the 3-2-1 rule

A **backup** is a copy of data in a safe location (cloud or external drive) so you can **restore** it after hardware damage, ransomware or accidental deletion.

The **3-2-1 backup rule** is a practical standard:

- **3** copies of the data (the original plus two backups).
- **2** different types of storage media (for example laptop disk + USB, or disk + cloud).
- **1** copy **off-site** (a different building or the cloud), so a fire or theft at home does not destroy every copy.

4. Firewall and antivirus software

A **firewall** is a security **barrier** that **monitors network traffic** and **blocks suspicious internet access**. Think of it as a checkpoint on the road between your device (or college lab) and the internet.

Antivirus software detects and removes harmful software (viruses, spyware and other malware) from the **local machine**.

Firewall $\approx$ guards the **door to the network**. Antivirus $\approx$ searches **inside the house** for intruders already present.

WORKED EXAMPLE

Worked example A — Caesar cipher, shift = 3. Encrypt the plaintext **SINDH**, then decrypt the result.

Write the alphabet and count three letters forward. Wrap after Z back to A.

Plain	S	I	N	D	H
+3	V	L	Q	G	K

Ciphertext = **VLQ GK**.

Decrypt: shift each letter **back** by 3: V → S, L → I, Q → N, G → D, K → H. You recover **SINDH**.

Spaces and punctuation are often left unchanged. The same key (3) is used both ways, so Caesar is **symmetric**.

Second check: **CAT** + 3 → C→F, A→D, T→W → **FDW**.

```
def caesar(text, shift):
    result = ""
    for ch in text:
        if ch.isalpha():
            base = ord("A") if ch.isupper() else ord("a")
            result += chr((ord(ch) - base + shift) % 26 + base)
        else:
            result += ch
    return result

print(caesar("SINDH", 3))
print(caesar("VLQ GK", -3))
print(caesar("CAT", 3))
```

OUTPUT

```
VLQ GK
SINDH
FDW
```

Worked example B — 3-2-1 for Fatima's Class XII project in Hyderabad.

1. **Copy 1 (original):** project folder on her laptop hard disk.
2. **Copy 2 (different media):** the same folder on a USB drive she keeps in her bag.
3. **Copy 3 (off-site):** uploaded to Google Drive (cloud servers are not in her house).

That is 3 copies, 2 media types (internal disk + USB, with cloud as the off-site copy), 1 off-site. When her brother deletes the laptop folder, she restores from USB or Drive.

Password check: Zk@2026#Pass has uppercase Z, lowercase letters, digits 2026, and symbols @ and #. fatima has none of those extras — reject it.

3 copies



2 media types



1 copy off-site

REMEMBER IT

Cipher = lock the message. Password = lock the door. 3-2-1 = spare keys in three places. Firewall = gate. Antivirus = indoor search. Caesar: shift letters; same key both ways (symmetric).

COMMON MISTAKE

Mistake 1: Shifting the wrong way, or forgetting wrap-around (X, Y, Z + 3 → A, B, C).

Mistake 2: Two copies on the same laptop (Documents and Desktop) and calling it 3-2-1. That is still one disk in one house.

Fix: Count letters on a written alphabet. For backups, different media and one copy that would survive a house fire.

اردو میں سمجھیں

ڈیٹا پروٹیکشن کا مطلب ہے معلومات کو غیر مجاز رسائی، غلط استعمال، نقصان یا حادثاتی حذف سے بچانا۔ انکرپشن پڑھے جانے والے متن کو سائفر ٹیکسٹ بنا دیتی ہے؛ سیزر سائفر میں ہر حرف کو مقررہ جگہوں سے آگے سرکایا جاتا ہے۔ مضبوط پاس ورڈ میں بڑے، چھوٹے حروف، عدد اور علامات ہوں۔ 1-2-3 بیک اپ: تین کاپیاں، دو قسم کے میڈیا، ایک کاپی باہر (آف سائٹ)۔ فائر وال نیٹ ورک ٹریفک روکتی ہے؛ اینٹی وائرس مقامی نقصان دہ سافٹ ویئر ہٹاتا ہے۔

CHECK YOURSELF

1. Encrypt PAK with a Caesar shift of 3.
2. What do the 3, the 2 and the 1 stand for in 3-2-1 backup?
3. In asymmetric encryption, which key encrypts and which key decrypts?

Answers: 1. SDN (P→S, A→D, K→N). 2. 3 copies, 2 different media, 1 off-site copy. 3. Public key encrypts; private key decrypts.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Encryption makes data:

- a. Unreadable without a key
- b. Faster to sort
- c. A primary key
- d. A pie

MCQ 2. A strong password is:

- a. Long, unique, not a word+123
- b. Ali123 for every site
- c. Written on the monitor
- d. OTP posted in the group

MCQ 3. Backup is:

- a. A second copy you can restore
- b. A stronger GPU
- c. A prettier UI
- d. A faster sort

MCQ 4. A firewall:

- a. Filters traffic in/out
- b. Encrypts Excel cells one-by-one always
- c. Sorts arrays
- d. Draws ER

MCQ 5. Ransomware defence includes:

- a. Offline/offline-ish backups
- b. Paying and hoping always
- c. No updates
- d. Shared admin

MCQ 6. Data protection is about:

- a. Confidentiality, integrity, availability (CIA idea)
- b. Only colours
- c. Only NAND
- d. Only sprints

Extra short questions

1. Four protection techniques from the notes.
2. Why unique passwords?
3. 3-2-1 backup idea (simple).
4. Firewall vs antivirus in one line.

Extra long question

1. A college lab is hit by ransomware. Using encryption, passwords, backup, firewall: what should already have been in place, and what do you do now (without paying if possible)?

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Encryption, strong passwords, backup, firewall (and updates, least privilege).
2. One leaked site should not open email+JazzCash.
3. More than one copy, not only on the same laptop.
4. Firewall: traffic gate. AV: looks for malware on the host (both useful).

Long

1. Before: backups offline, least admin, updates, firewall. Now: isolate PCs, restore from backup, change passwords, report. Paying is not a syllabus requirement to endorse.

Next lecture: 5.4 Security Threats and Mitigation Techniques